

Atividade 21

Print da atividade 11.1: Identificação de interfaces de rede e verificação de endereçamento IP via comando ip address no Kali Linux

```
(root@Cleydson)-[~]
# arp -a
? (192.168.15.109) em a8:e6:21:2d:62:64 [ether] em wlan0
GPT-2741GNAC-N2-SV.GPT-2741GNAC-N2-SV (192.168.15.1) em 84:0b:bb:6c:08:a8 [ether] em wlan0
? (192.168.15.58) em 6c:ff:ce:85:2d:69 [ether] em wlan0
? (192.168.15.16) em c4:57:6e:6e:fa:9c [ether] em wlan0

(root@Cleydson)-[~]
# sudo ip -s -s neigh flush all
192.168.15.109 dev wlan0 lladdr a8:e6:21:2d:62:64 used 206/159/129probes 3 STALE
192.168.15.1 dev wlan0 lladdr 84:0b:bb:6c:08:a8 ref 1 used 22/22/22probes 1 REACHABLE
192.168.15.58 dev wlan0 lladdr 6c:ff:ce:85:2d:69 used 209/209/185probes 1 STALE
192.168.15.16 dev wlan0 lladdr c4:57:6e:6e:fa:9c used 5444/5444/5410probes 1 STALE
fe80::860b:bbff:fe6c:8a8 dev wlan0 lladdr 84:0b:bb:6c:08:a8 router ref 1 used 470/1/470probes 1 REACHABLE

*** Round 1, deleting 5 entries ***
*** Flush is complete after 1 round ***

(root@Cleydson)-[~]
# arp -a

(root@Cleydson)-[~]
# ping -c 4 google.com
PING google.com (2800:3f0:4001:800::200e) 56 data bytes
64 bytes from 2800:3f0:4001:800::200e: icmp_seq=1 ttl=253 time=11.6 ms
64 bytes from 2800:3f0:4001:800::200e: icmp_seq=2 ttl=253 time=30.9 ms
64 bytes from 2800:3f0:4001:800::200e: icmp_seq=3 ttl=253 time=14.8 ms
64 bytes from 2800:3f0:4001:800::200e: icmp_seq=4 ttl=253 time=17.3 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 11.574/18.639/30.857/7.342 ms

(root@Cleydson)-[~]
# arp -a
GPT-2741GNAC-N2-SV.GPT-2741GNAC-N2-SV (192.168.15.1) em 84:0b:bb:6c:08:a8 [ether] em wlan0
? (192.168.15.58) em 6c:ff:ce:85:2d:69 [ether] em wlan0

(root@Cleydson)-[~]
#
```

Print da atividade 11.2: Alteração manual do endereço IP e máscara de sub-rede para isolamento em ambiente de laboratório

```
(root@Cleydson)-[~]
# ping -c 2 10.0.0.21
PING 10.0.0.21 (10.0.0.21) 56(84) bytes of data.
64 bytes from 10.0.0.21: icmp_seq=1 ttl=64 time=0.048 ms
64 bytes from 10.0.0.21: icmp_seq=2 ttl=64 time=0.095 ms

— 10.0.0.21 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1008ms
rtt min/avg/max/mdev = 0.048/0.071/0.095/0.023 ms

(root@Cleydson)-[~]
# ssh seu_usuario@10.0.0.22
ssh: connect to host 10.0.0.22 port 22: Connection refused

(root@Cleydson)-[~]
# sudo systemctl start ssh

(root@Cleydson)-[~]
# ssh root@10.0.0.22
The authenticity of host '10.0.0.22 (10.0.0.22)' can't be established.
ED25519 key fingerprint is: SHA256:hJvyl7375zK/zztD6L/57zbt4uryedNs1cv9/0aIQtI
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.0.22' (ED25519) to the list of known hosts.
(root@10.0.0.22) Verification code:
(root@10.0.0.22) Password:
(root@10.0.0.22) Verification code: █
```

Print da atividade 11.3: Uso da ferramenta macchanger para ocultação e troca do endereço físico (MAC) original da interface

```

(root@Cleydson)-[~]
# sudo ip addr del 10.0.0.21/24 dev wlan0
sudo ip addr del 10.0.0.22/24 dev wlan0
sudo systemctl stop ssh

(root@Cleydson)-[~]
# ip link show wlan0
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT group default qlen 1000
    link/ether c0:35:32:a7:42:7d brd ff:ff:ff:ff:ff:ff

(root@Cleydson)-[~]
# sudo ip link set wlan0 down

(root@Cleydson)-[~]
# sudo ip link set dev wlan0 address 08:00:27:8b:c8:b5

(root@Cleydson)-[~]
# sudo ip link set wlan0 up

(root@Cleydson)-[~]
# sudo ip link set wlan0 down
sudo ip link set dev wlan0 address SEU_MAC_ORIGINAL_AQUI
sudo ip link set wlan0 up
ip link show wlan0
"SEU_MAC_ORIGINAL_AQUI" is invalid lladdr.
3: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DORMANT group default qlen 1000
    link/ether 08:00:27:8b:c8:b5 brd ff:ff:ff:ff:ff:ff permaddr c0:35:32:a7:42:7d

(root@Cleydson)-[~]
# sudo ip link set wlan0 down

(root@Cleydson)-[~]
# sudo ip link set dev wlan0 address c0:35:32:a7:42:7d

(root@Cleydson)-[~]
# sudo ip link set wlan0 up

(root@Cleydson)-[~]
# ip link show wlan0
3: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN mode DORMANT group default qlen 1000
    link/ether c0:35:32:a7:42:7d brd ff:ff:ff:ff:ff:ff

(root@Cleydson)-[~]
#

```

Print da atividade 11.4: Aplicação de QoS para limitação da largura de banda de download e controle de tráfego

```
root@Cleydson: ~  
Sessão  Ações  Editar  Exibir  Ajuda  
(root@Cleydson)-[~]  
# sudo tc qdisc del dev wlan0 root  
Error: Cannot delete qdisc with handle of zero.  
(root@Cleydson)-[~]  
# sudo tc qdisc add dev wlan0 root tbf rate 1mbit burst 10kb latency 50ms  
(root@Cleydson)-[~]  
# tc qdisc show dev wlan0  
qdisc tbf 8001: root refcnt 2 rate 1Mbit burst 10Kb lat 50ms  
(root@Cleydson)-[~]  
#
```

Print da atividade 11.5: Restrição da velocidade de upload para garantir a disponibilidade de serviços críticos na rede local

